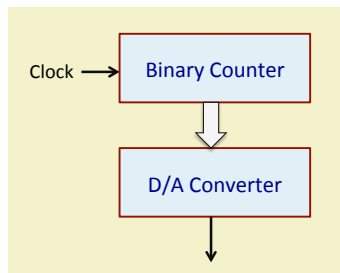


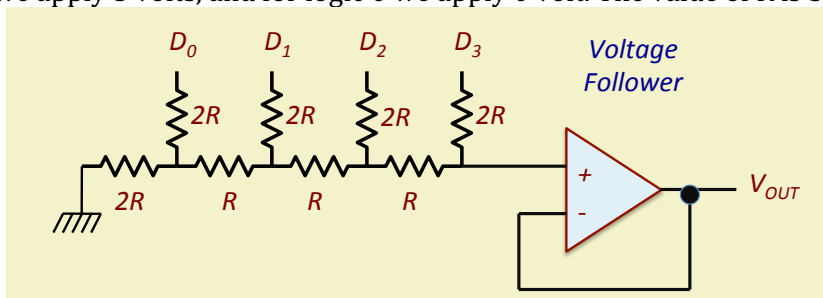
## Module 9 :: D/A and A/D Converters

### Practice Problems

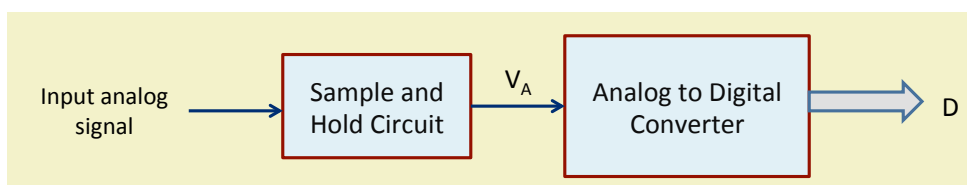
3. Consider a 3-bit D/A converter with full-scale output voltage of 14 volts. The D/A converter is fed with the outputs  $Q_2Q_1Q_0$  of a 3-bit binary counter as shown in the diagram below. Suppose accidentally the  $Q_2$  line is permanently connected to logic 1. As the counter counts through 0, 1, 2, 3, 4, 5, 6 and 7, what will be the output voltages observed?



- a) 6V, 10V, 12V, 14V, 8V, 10V, 12V, 14V  
 b) 4V, 6V, 4V, 4V, 12V, 14V, 12V, 14V  
 c) 0V, 2V, 0V, 2V, 12V, 14V, 12V, 14V  
 d) 14V, 12V, 10V, 8V, 6V, 4V, 2V, 0V
2. Consider a 4-bit resistive-ladder type digital-to-analog converter as shown below, where ( $D_3 D_2 D_1 D_0$ ) is the digital input, and  $V_{OUT}$  is the analog output. For an applied logic 1 to any of the digital inputs, we apply 5 volts, and for logic 0 we apply 0 volt. The value of  $R$  is 5 Kohm.



- a) When we apply 1101 to the input, what will be the output voltage?  
 b) When we apply 0101 to the input, what will be the output voltage?
3. A band-limited analog signal is fed to a 12-bit analog-to-digital converter through a sample-and-hold circuit, as shown in the following diagram. Assume that the input analog signal contains the frequency components 2 KHz, 7.5 KHz and 15 KHz.



- a) What is the minimum sampling rate so that the signal can be perfectly reconstructed at the receiving end?

- b) If the A/D converter is constructed as a flash-type converter, what will be the number of voltage comparators required?
  - c) If the A/D converter is constructed as successive-approximation type, what will be the number of clock cycles required for each conversion?
4. Arrange the following A/D converter types according to their speed of conversion with proper justifications: counter-type, flash-type, tracking-type, successive-approximation-type.